



Guide for the Tenure-Track Job Market in Computer/Information Sciences

This guide is part of my [advice pages](#).

Read This Part First

I wrote this guide for graduate students, postdocs, and others who seek a faculty position with all of the following parameters:

- Tenure-track (hiring at assistant professor or open rank);
- Research-oriented (i.e., with at least as much emphasis on research as teaching, though both are obligatory for tenure-track faculty);
- In computer science or information science ("CS/IS" below for brevity); and
- At a university in the USA.

Judging by the quantity of ads that appear every year, this is one of the most robust (or perhaps least anemic) sectors of the academic job market. Still, openings are very competitive, and academic job-seekers should also develop fall-back plans that they are comfortable with. Positions in government and industry are typically easier to get, and they provide different advantages. Within academia, teaching-oriented faculty positions and non-tenure track research positions also exist.

If your parameters differ from the above list, the applicability of this guide may vary. For example, hiring committees in some disciplines perform screening interviews at conferences, but in CS/IS screening interviews tend to be via teleconference. Interviews for teaching-oriented positions often require a teaching demo, but research-oriented positions do not. Universities in other countries may hire at a different time of the year than universities in the USA.

A PhD is a firm requirement for obtaining a faculty position with these parameters. If you don't have a PhD and you're interested in my [research](#), take a look at the [Guide for Joining My Lab](#). In the [Basics](#) section I answer some questions on what getting a PhD is like and I suggest how to self-evaluate whether you should or shouldn't pursue one.

Note that the terms *research-track* (often *research professor*) and *teaching-track* (*teaching professor*) represent other kinds of positions not covered by this guide. See my [Guide for Interacting With Faculty](#) for details. Tenure-track positions are highly sought after because of the stability, the balance of activities, and the prestige. I list some of the things I enjoy about being a professor in [My Academic Journey](#). However, the other tracks may let you focus on the aspects of being a professor that you most enjoy.



detailed records.) If you plan to apply for academic positions, you should consult additional resources like the archives of [The Professor Is In](#). Also, the [Taulbee Survey](#) provides excellent data on the state of the CS/IS tenure-track job market. Additionally, you should speak with your PhD advisor and faculty members in your department who recently served on search committees.

This guide contains my firsthand observations from several years on the academic job market. The first was the 2012-2013 cycle, and the most recent was the 2017-2018 cycle. The vast majority of the experiences I describe were from applying for positions that fit the parameters I listed above. I've also served on search committees since becoming faculty, and I include some observations from the employer's perspective.

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Seashells on a beach on Isle of Raasay, Scotland.

Deciding Whether to Apply

For some people, the appeal of an academic position is tremendous: job security, flexible work hours, variation in activities, frequent contact with the research community, working with students, travel opportunities, and a mission to serve the public good. There are also drawbacks: pre-tenure uncertainty, long work hours, vague performance expectations, and frequent rejection of papers and funding proposals.

Only you can decide whether to apply for faculty positions, but here are some positive signs:

- **You enjoy working with students:** You will teach courses, typically at both the undergraduate and graduate levels. You will have research advisees, and your success as a researcher will depend in part upon their success.
- **You're OK with a job that requires a lot of human interaction:** Students in your courses, advisees, colleagues in your academic unit, and collaborators elsewhere will occupy much of your time. Successful faculty are distributed across the spectrum from introversion to extroversion, but there's no way around the need to work with people.
- **You're comfortable self-managing your work:** Only rarely will your supervisor (typically a department head, head of school, or dean) give you instructions for your work. Tenure-stream faculty are expected to be self-driven and to find the resources they need for success on their own.
- **The university environment appeals to you:** Research universities differ from private companies, nonprofit labs, or other work environments. The centrality of students and the emphases on grants and publishing are some of the distinguishing features.
- **You're comfortable with being a research manager:** You will be responsible for interviewing and hiring graduate and undergraduate research assistants. Then, you must guide their performance and secure resources for them, as tenure-stream faculty in STEM disciplines tend to rely on students to perform many hands-on research activities. Occasionally you may have to make difficult decisions about a student's future with your lab.



life. They may have advice and opinions, but ultimately the decision is yours.

- Your greatest motivation is social status: Satisfaction chiefly from social status is unlikely to last long once you are faced with the challenges of the job.
- You're motivated by money: Although CS/IS tenure-stream professors at R1s are relatively well-paid faculty, CS/IS PhDs have more lucrative career options in industry.

Preparing Your Materials

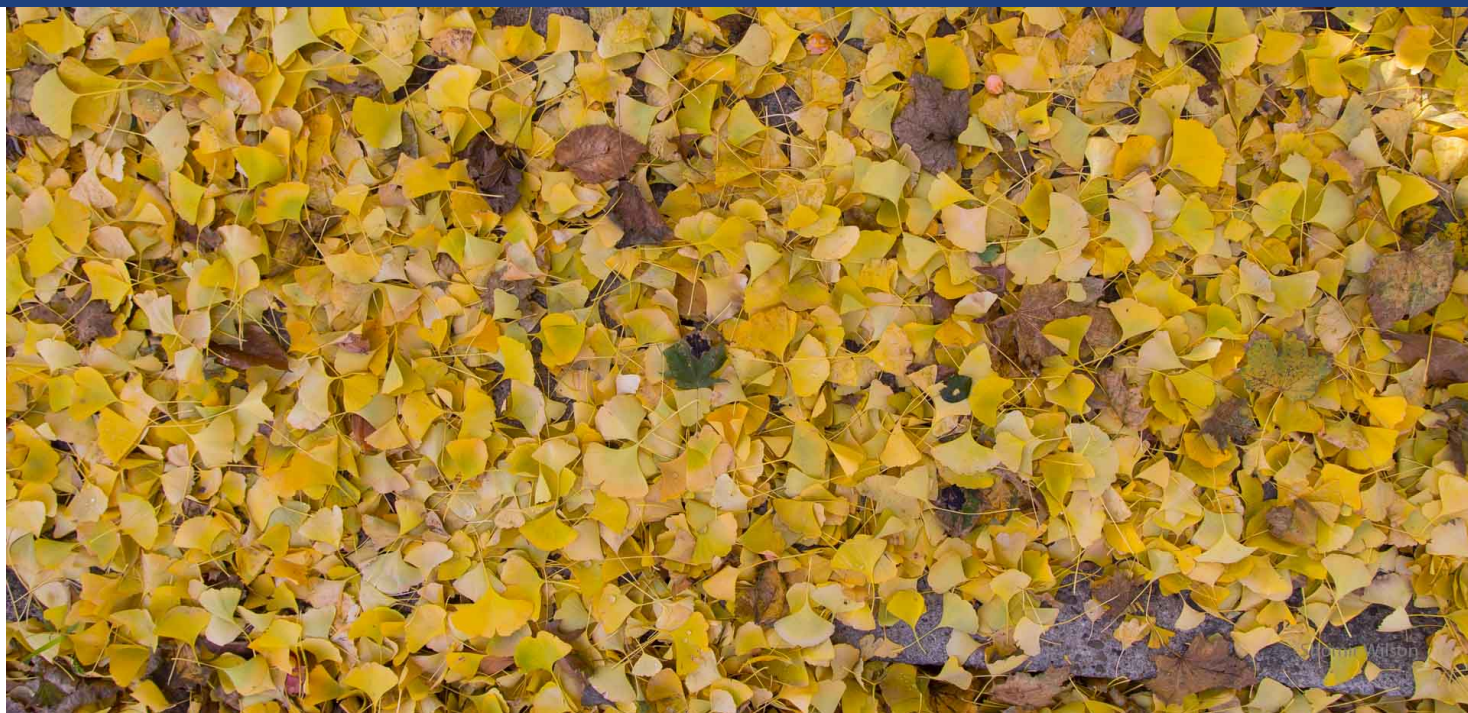
An application for a faculty position typically consists of the following:

- **Basic Information:** Your name, contact information, where and when you received your degrees, etc. This is the easy part.
- **Curriculum Vitae (CV):** Members of the hiring committee sometimes read this before the cover letter. [This page](#) contains a sample CV that I created for PhD students, complete with LaTeX source code. You can also refer to [my current CV](#).
- **Cover Letter:** It should be one or two pages, describing your interest in the position and your qualifications in both research and teaching. Some customization is appropriate for each opening: you should name the academic unit and the university, as well as any faculty there whom you see strong collaborative potential with. When I knew the name of the head of the search committee, I addressed them in the salutation.
- **Research Statement:** Describe your research accomplishments and what you will do next. Your future work should strike a balance between building upon your strengths and breaking new ground in your discipline. My research statement varied in length by year, but the optimal length felt like 2-3 pages. I included citations and abbreviated references, mostly to my publications but also to a few others.
- **Teaching Statement:** Describe your teaching experiences, your strengths working with students, and a mix of undergraduate and graduate courses that you can teach. Hiring committees for research-oriented positions pay less attention to this than to the research statement, but you still must establish that you are a competent teacher and you are capable of reflection upon your skills. 1-2 pages is sufficient.
- **Sample Publications:** Applications typically require three. Choose a balance between your most recent work and your most impactful work.
- **Contact Information for References Who Can Provide Letters of Recommendation:** Applications typically require three, but a few require four. Sometimes references are contacted as soon as you submit the application, and sometimes they are contacted at a later stage. It's safest to assume they will be contacted immediately.

Some applications require a diversity statement. If you are asked to provide one, you should explain your experiences with supporting diversity and how you will encourage a diverse, inclusive environment as a faculty member. 1-2 pages is sufficient.

Start working on your materials several months before you apply for faculty positions. Get feedback from your advisor, from other professors you feel comfortable asking, and from your fellow students. My [Guide for Scholarly Writing](#) is applicable to some of these materials.

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Ginkgo leaves in the fall in Pittsburgh, Pennsylvania.

Applying for Openings

Fall is when nearly all of the job ads are posted. They typically appear on [CRA Jobs](#), [AcademicJobsOnline](#), and [HigherEdJobs](#). A few appear in [HERC](#) and [ChronicleVitae](#). I recommend using an RSS reader (like [Feedly](#)) to simplify checking all these sources for new ads. If you are very interested in working in an academic unit at a particular university, you may check the academic unit's website or the university's job site. However, take note of the posting dates for the ads you find in those places, as there is a chance that they are from a previous hiring cycle.

For openings I applied to in Fall 2017, here is an approximate breakdown of when ads were posted and when applications were due:

Month	% Postings	% Deadlines
August	10%	-
September	30%	10%
October	40%	20%
November	10%	50%
December	10%	20%

A few ads had a posting date or a deadline outside of this range. Also, a small number of ads did not list deadlines because they sought applications as soon as possible.

Search committees' procedures vary widely. Some read applications as they come in, and some do not look at them until the submission deadline passes. (Some also read applications submitted after the deadline passes, so don't disregard openings with deadlines you missed.) Committees that request reference letters when an application is submitted may feel more or less strongly about needing an applicant's letters before considering them for an interview.



record of how you presented yourself.

A tense wait begins after you submit applications. It may end with your first invitation to interview, but given the competitiveness of academic positions, it's also possible that you will not receive any invitations. Spend the time developing alternate prospects, such as postdoc positions and jobs in government or industry.

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First Contact

I kept track of how much time elapsed between an application deadline and "first contact", i.e., either an invitation for a remote screening interview or an invitation to interview on campus without screening first. The mean time was 31 days. The median was 30 days, the minimum -23, and the maximum 133. 80% of all first contacts were no more than 60 days after the submission deadline.

85% of first contacts happened between the first week of December and the last week of March, with roughly even distribution across that time period. The earliest was in September. The latest was in May, from a non-US university I interviewed with.

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Ice in a creek in rural southeast Indiana.

Screening Interviews



For videoconferences I wore a dress shirt with a tie and a suit jacket or sport coat. I often wore these things for telephone interviews as well, to get into an interviewing frame of mind.

For videoconferences, make sure that the background behind you is not distracting for your audience, and check that the lighting is appropriate. Also, use a video device that you do not have to hold in your hand, as shaky video will distract your audience.

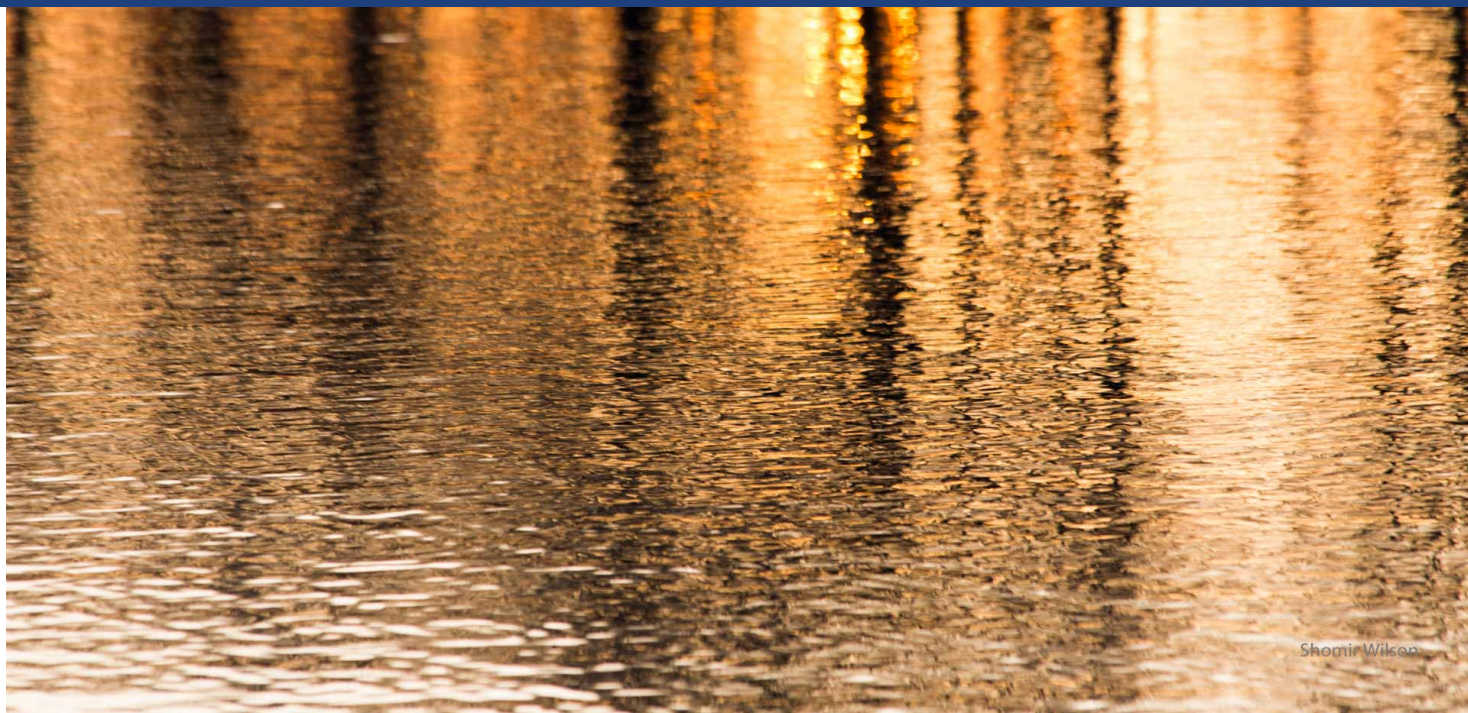
Typically your audience will be 2-5 faculty members who are on the search committee. (On a few occasions I spoke with just one faculty member, and on one occasion I spoke with ten.) Some common questions to prepare for:

- **Why are you interested in joining the academic unit?** Your answer should be specific to the academic unit and the university; simply wanting a faculty job anywhere is not enough. Your answer should be mostly professional in nature, although if there are lifestyle reasons (i.e., a fondness for the university or its surroundings) then briefly mentioning them toward the end of your answer is OK.
- **Whom at the university would you collaborate with on research?** Have at least four tenured or tenure-track faculty in mind. (Interviewers typically ask for three, but on one occasion I was asked for four.) They should be mostly in the academic unit that is hiring, but it's fine if one or two of them are in a different unit at the same university.
- **What are your teaching interests, or what is your teaching philosophy?** The former question is about what classes you can teach, and the latter is about your approach to teaching.
- **What questions do you have for the search committee?** At this stage, keep your questions simple and general. I often asked about the department's planned trajectory over the next five years. I always ended by asking about the hiring committee's next steps and when I should expect to hear from them again.

For a videoconference, consider standing up, provided that you can arrange your laptop or other device at an appropriate height to capture your head and shoulders. I found it helpful to tape my notes to the wall behind the laptop, so I could consult them while speaking.

A note for my PhD advisees: If you need a quiet, professional-looking place for a videoconference interview, ask me about using my office. I'll step out during the call.

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Sunset reflections in Montreal, Canada.

Campus Interviews

When a screening interview preceded an invitation to a campus interview, the mean time between them (i.e., between the screening interview and the campus invitation) was 15 days. The median was 10 days, the minimum one, and the maximum 44. Three-quarters of all campus invitations came within 14 days after the screening interview.

69% of my campus interviews were between the first week of December and the last week of March. The earliest was in October and the latest was in June, at a non-US university.

On-campus interviews are one to two days long. **Throughout all activities with your hosts, behave as if you are being evaluated.** Everyone on your schedule, including students, will be asked to send their comments about you to the hiring committee. However, if you are nervous about a campus interview, it may help to remember that a healthy academic unit is supportive of faculty candidates and makes an effort to find the best in them. Hiring committees want to show that they've done their job well. Administrators want to show that their academic units attract top talent. Junior faculty remember the high stakes of their interviews, and they are quietly sympathetic. Senior faculty think about the department's future, and they want it to be prosperous. Your success as a candidate is desirable for all of these parties.

Here are the typical components of a campus interview:

- **Many one-on-one meetings with faculty:** These tend to be a half-hour in duration, and they will occupy most of your schedule. Some faculty will have many questions for you, and some will expect you to ask all the questions. Some may spend the entire time talking, leaving you with no time to speak. Keep an eye on the time. If there is no clock in sight, asking "How are we on time?" is a polite way to check and an implicit reminder that your schedule is a shared responsibility.



- **Meetings with administrators:** These include the department head and either the dean or an associate dean. Conversations with them tend to be less focused on your research results and more focused on the resources you need to perform your research.
- **Your job talk:** This tends to be an hour long, including 10-15 minutes for questions at the end. A good job talk is engaging for a wide audience of computer scientists and information scientists. It should demonstrate the strength of your research portfolio and the viability of your future research plans. Practicing your talk is important. Prior to interviewing, give your talk to an audience in your home academic unit.
- **A meeting with students:** Typically you meet with a small group of graduate students, although some undergraduates may be invited. Speaking with them is an excellent opportunity to gauge the mood and trajectory of the academic unit.

Here are some less common items:

- **A meeting with the hiring committee:** Sometimes my schedule included a special session to allow the hiring committee to ask their questions as a group, but more often the committee members had half-hour sessions just like other faculty.
- **A meeting with staff:** A few times I had meetings with staff to discuss my needs for research support and what the academic unit could offer.
- **A tour of the area:** A few times the academic unit arranged for a realtor to drive me around, pointing out different neighborhoods and local amenities. This gave me some basis to feel a connection with the area, and I wished more academic units had done this.

The campus interview is a two-way evaluation. Look for signs that an academic unit is a productive, properly functional place to work. These include:

- **Strong job talk attendance:** People should want to see you speak.
- **Enthusiastic faculty:** Junior faculty should be optimistic about their careers, and senior faculty should be reassuring and supportive.
- **Consideration for your personal interests:** No one should ask you illegal questions (e.g., about your marital status or whether you have children), but it's a good sign if faculty and staff are curious about ways they can relate to you.
- **Enthusiastic students:** It's a good sign if they are happy with their classes and their degree requirements, even if they have some criticisms. PhD students should feel confident that they will remain funded as RAs or TAs or via a fellowship until they finish their degrees.

As a job candidate, getting a well-rounded perspective on an academic unit can be difficult. Keep in mind some strategies and cautions when speaking with current faculty:

- Asking directly whether the department is collaborative or interdisciplinary is unlikely to produce the information you want, as interviewers will feel obligated to answer positively. Instead, it's better to ask specifically about an interviewer's collaborations with colleagues in the academic unit and with faculty in other parts of the university.
- If you want to ask about teaching load, be aware that simple answers are probably misleading. An unqualified answer (e.g., "it's three per year") might be the default pre-tenure teaching load, the teaching load for faculty who meet certain thresholds for research activity, the university's default teaching load (and it's possible no one in the department has it), the post-tenure default teaching load, the special teaching load of an endowed chair you're speaking with, or another number.
- The concerns of junior faculty are the most relevant to your evaluation of an university (since you would soon become one of them), but junior faculty can be especially reluctant to criticize their employer. Asking a question like



- One way to gauge the vitality or morale of senior faculty is to ask how they expect the department to change over the next five years. New degree programs, growth in a certain research area, new facilities, or other new initiatives are good signs. In contrast, stagnation is likely to subtract from morale.

Some warning signs (one instance of any of these might be a fluke, but more than one is a cause for concern):

- Curmudgeonly interviewers: In one-on-one conversations with faculty, some may be heavily critical of their academic unit. You should never express agreement with a curmudgeon's criticisms, but you can carefully ask questions to explore their perspective.
- No-show interviewers without explanations: People on your schedule should care about meeting you.
- Unhappy students: When you meet students, ask them to name some things they like about the department. If they cannot, or if they complain excessively, the faculty are likely to be unhappy too.
- Arguments during job talk Q&A: Faculty should not be hostile toward your research or toward each other.

I dressed in a suit and tie, and I carried my laptop in a gray messenger bag.

Send individual thank you emails to your interviewers a day or two later. This reinforces their memory of you.

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Rock face in Point Lobos State Nature Reserve, California.

Understanding Rejection



In some cases, not being interviewed may feel like a surprise. It's possible to be passed over for an interview even if you have strong connections with the faculty in an academic unit, you know someone on the hiring committee who likes your work, and you have a stellar portfolio that precisely fits the ad. Remember it's common for job openings in this space to be swamped with more strong candidates than the hiring committee can interview, forcing them to make arbitrary-seeming distinctions while ranking candidates. I call this *criteria saturation* in [Thoughts on Competition](#).

Faculty searches sometimes fail—i.e., they hire no one—even without interviewing all the strong candidates. If none of the interviewees who received offers accept them, there might not be enough time left in spring semester to consider more applications or interview more candidates. (Hiring committees typically must finish all their activities before exam week and summer break, when their members become much less available.) Other reasons for a failed search include lack of agreement on which applicants to interview next or withdrawal of an administrator's support for hiring.

If you have close connections in an academic unit that didn't invite you to interview, it might be tempting to ask them why. However, any feedback you gather is subject to constraints. It's possible no root cause led to your application being rejected. It's also possible no one involved in hiring has the full picture of why decisions were made, and then comparing your applications to others' will have limited value. If there's another opening next year, many things are likely to change: the academic unit's hiring needs, the committee members (along with their opinions on hiring), and the other applications yours will compete with.

Under these circumstances—rejections that are difficult to interpret and the limitations of root-cause framing—having several trusted mentors is helpful. It's possible that satisfying explanations for rejection won't exist, but if your personal circumstances permit enough time, you can improve your portfolio and try again next year.

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Offers

Totalling across all of my years on the academic job market, I received four hard offers (i.e., official letters) and one soft offer (an informal notification of intent to make an offer). Three of my hard offers arrived between 3-5 weeks after I interviewed, and the fourth was an unusual situation that took much longer. However, note that these durations included pre-offer conversations initiated by the university to determine what I would want in an offer, indicating an intent to make an offer but not yet making my selection official.

Academic units vary in how they decide which candidates should receive offers. The search committee may play a role in choosing offer recipients, or its role may effectively end after the candidates interview. Typically the faculty discuss candidates over an email list and/or during a department meeting. Afterward a poll is held, either to rank candidates or to express attitudes toward each one, and then a department head or dean makes final decisions.

Unless you have an offer from a different institution, you gain little by prodding an academic unit for updates on your candidacy. Just be patient. (This applies to all earlier stages of the process, too.) However, if you receive an offer then it's reasonable to let your other top choices know, in case they want to compete for you.

Check out [TPII's advice on negotiating](#), which is largely applicable within the parameters of this guide. For a [startup package](#), it's typical to receive some funding to support PhD students, travel funds to go to conferences, equipment funds, and some summer salary. Moving expenses and a few teaching releases are also typical in an offer, though they may be technically separate from the startup package.



The knowledge you gained while interviewing and negotiating could be valuable to others. Consider how you can share what you learned with your peers and with your broader academic communities.

If you accept an offer, you might want to read [Tenure Worth Wanting](#), [Guide to Professorspeak](#) and [Inglorious Proposals](#).

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Tree bark in Pittsburgh, Pennsylvania.

Moving Between Universities

Tenure-line faculty sometimes move from one university to another. Although moves are rarer for tenure-line faculty than for people in other jobs, they are still a normal part of a career in academia. Faculty who return to the academic job market might be unsatisfied with their employer, want to live in a different place, or have personal constraints that require them to relocate. In some cases a faculty member might be satisfied where they are but they see an even more attractive opportunity elsewhere. Still, judging by the questions I receive about this topic, many people are unaware that it's possible to move or afraid they will be stuck one place forever.

I moved after my first two years as faculty because I discovered my first academic unit wasn't a good fit for me. I know many faculty who have moved once, and I know a few faculty who have moved twice or more (i.e., they worked at three or more universities). I'm aware of at least two occasions when a faculty member who interviewed me was on the academic job market too, though I didn't know until a few months later, when they moved. It's likely other faculty who interviewed me were also looking around.

Some things to remember if you decide to return to the academic job market after landing a position:



is a polite way to imply that you're motivated to move while pivoting to discuss the merits of your hosts' institution.

- Seniority is only one dimension of your candidacy, which will be evaluated on other dimensions too. You will compete with graduating PhD students; don't be surprised when hiring committees make offers to them instead of you, and try not to feel bad about it.
- Ideally you should get reference letters from senior colleagues at your current institution who have seen you work as faculty. However, if that's not possible, try asking faculty whom you overlapped with at your current institution but who recently departed. They should understand from personal experience that making a career move is normal.
- Avoid asking interviewers about credit toward tenure. Like most other offer parameters, it's best to wait until you receive an offer, to avoid introducing doubt about your willingness to move.

Pre-tenure faculty who move between universities are sometimes offered credit toward their [tenure clock](#) for years they've already worked. For example, I was pre-tenure faculty for two years before I moved to another university, and I had options to request zero, one, or two years' reduction to the six-year tenure clock. (I requested zero, because I wanted as much time as possible to prepare.) There also exist cases when moving faculty are not allowed credit for their pre-move years. Regardless, credit for time spent in non-tenure-line positions is unlikely, and moving from a pre-tenure position to one with tenure on arrival is rare or impossible.

Tenured faculty who move to a new university face slightly different circumstances. Possible scenarios include tenure on arrival, starting without tenure and applying for it during the first year, and waiting longer to apply. The faculty member has little control over which of those happen, which may seem precarious, but universities are unlikely to spend the time and money to hire a mid-career or senior tenure-line faculty member without intending to keep them.

Reading a university's tenure policies is a good idea, but whenever they contain flexibility, individual academic units may apply that flexibility differently. Because of limited public information, broad claims about how certain universities handle transferring faculty are unlikely to have sufficient factual basis, as are broad claims about [R1s](#).

I write more about considering whether to make a move under "[Remember It's a Job](#)" in [Tenure Worth Wanting](#).

If You Don't Receive an Offer

This section contains the most important advice in this guide. It is also the most difficult advice to follow.

If you did not receive any interview invitations, recognize that it was not a judgement on your personal worth. It might not even qualify as a judgement on your professional portfolio: departmental fit, hiring priorities, trendiness, and a dozen other variables exclude applicants according to a pattern that feels random at best. If you interviewed but did not receive an offer, recognize that you were competing in the tightest segment of the job market for CS/IS graduates. **Even being a contender in this space is an achievement.**

Read my [Thoughts on Failure](#) for some perspective. I describe how it took me four years on the academic job market to receive my first offer.

If you've followed my encouragement to develop alternate prospects, it's time to take one of them. Regardless of the direction you choose, you may consider applying again the following year. If you're leaning toward a second try, have conversations with your advisor, with PhD-holding colleagues inside and outside of academia, and with friends who understand your motivations.